

2. Positive definite matrices

② A matrix is positive semidefinite if it is symmetric ($A^T = A$) and if $x^T A x \geq 0 \forall x \in \mathbb{R}^n$.
 ✓ Therefore, to demonstrate $A = z z^T$ is PSD, we will demonstrate both assertions.

A is symmetric:

$$A = z z^T = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix} \begin{bmatrix} z_1 & z_2 & \dots & z_n \end{bmatrix} = \begin{bmatrix} z_1^2 & z_1 z_2 & \dots & z_1 z_n \\ z_2 z_1 & z_2^2 & & \vdots \\ \vdots & & \ddots & \\ z_n z_1 & \dots & & z_n^2 \end{bmatrix}$$

This shows that in the matrix $A_{ij} = A_{ji} \quad \forall i, j \in [1, n]$
 Therefore, the matrix is symmetric and $A = A^T$

Quadratic form of A is always ≥ 0 :

Start by computing the quadratic form:

$$x^T A x = x^T z z^T x. \quad \text{For associativity of matrix multiplication: } x^T z z^T x = (x^T z)(z^T x)$$

when $x^T z = (z^T x)^T$. Therefore, letting $y = z^T x$, $y \in \mathbb{R}$ and $x^T A x = y^2$ which is always ≥ 0 . Therefore $x^T A x \geq 0 \quad \forall x \in \mathbb{R}^n$

We could also write: $y = \sum_{i=1}^n x_i z_i$ and $x^T A x = \left(\sum_{i=1}^n x_i z_i \right)^2 \geq 0$

could simply verify that $A^T = A$
 $A^T = (z z^T)^T = z z^T = A$

⑥ The nullspace of a matrix is defined as $N(A) = \{x \in \mathbb{R}^n : Ax = 0\}$

✓ In our case, this is the set of all vectors such that $z z^T x = 0$ (expanding A)
 The condition is verified if $x = \vec{0}$ (x is the null vector),

$$z(z^T x) = 0 \Rightarrow z^T x = 0, \text{ i.e. } x \text{ and } z \text{ are orthogonal.}$$

The nullspace of A is, therefore, the set of all the vectors orthogonal to z , and the null vector.

Regarding the ~~rank~~ of A , each A_i can be expressed as $A_i = z_i \cdot z$ (see above the structure of A)

Therefore, each column of A can be expressed as z scaled version of a single vector.

Assume at least one $z_i \neq 0$. Then $z = \frac{1}{z_i} A_i$. All vectors $A_j = \frac{z_j}{z_i} A_i$

All vectors can be expressed as a linear combination of other vectors, namely using $\frac{z_j}{z_i}$ as coefficient for A_i , and 0 otherwise. Only 1 column is linearly independent $\Rightarrow R(A) = 1$

⑦ If $A \in \mathbb{R}^n$ and $A^T = A$, it was already illustrated that A can be diagonalized as

$$A = U \Lambda U^T, \text{ with } U \text{ being a matrix of normalized eigenvectors of } A.$$

In addition, since Λ is diagonal, there exists a matrix $\Lambda^{1/2}$ such that: $\Lambda^{1/2} (\Lambda^{1/2})^T = \Lambda$

Therefore, the quadratic form of BAB^T is:

$$x^T BAB^T x = x^T B U \Lambda U^T x = x^T B U \Lambda^{1/2} (\Lambda^{1/2})^T U^T B^T x$$

It can be shown that $(U^T B U \Lambda^{1/2})^T = (\Lambda^{1/2})^T U^T B^T x$. Therefore, ~~then~~ $J = x^T B U \Lambda^{1/2}$,
 the quadratic form is $J J^T \geq 0$, independently of B

Similarly, it can be demonstrated that BAB^T is symmetric.

$$(BAB^T)^T = B A^T B^T \quad \text{but } A^T = A \text{ because } A \text{ is symmetric}$$

$$(BAB^T)^T = BAB^T \quad \text{and also } BAB^T \text{ is symmetric}$$